

n (odd) Voters:

For a supporter of A voting for B is weakly dominated.

Proof: i) It does not make any diff to the outcome so the voter is indifferent b/w two actions

ii) It makes A the winner compared to the case when B is the winner.

all NE?

1) win margin = 1 all voters of the winner like the winner

2) All other profiles (margin = 2, ...)

Game (Collective Decision Making):

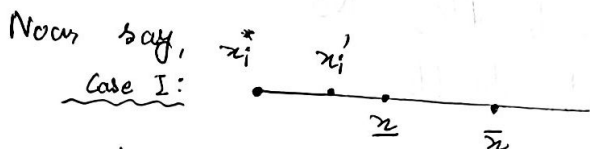
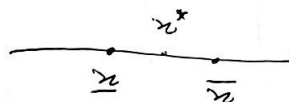
$i = 1, 2, \dots, n$ ($n = \text{odd}$)

i announces $x_i \rightarrow (x_1, \dots, x_n) = a$, $\text{Med}(a) = x^*$

minimize, $|x_i^* - x^*|$

Now, here, $x_i = x_i^*$ weakly dominates all other x_i ; prove.

Keeping i aside there are $n-1$ announcements. Let two middle most observations be $\underline{x}$ and $\bar{x}$ ie,



Case I:

and so on for other cases.

we choose x_i^* med = $\underline{x}$
and choose x_i' med = $\bar{x}$

so, given, $x_i^* < x_i'$

	$x_i^* < x_i' < \underline{x}$	$x_i^* < \underline{x} < x_i' < \bar{x}$	$\bar{x} < x_i'$	$\underline{x} < x_i^* < x_i' < \bar{x}$	$\underline{x} < x_i^* < \bar{x} < x_i'$	$\bar{x} < x_i^* < x_i'$
Med if $x_i^* = x_i$	$\underline{x}$	$\underline{x}$	$\underline{x}$	x_i^*	x_i^*	$\bar{x}$
Med if $x_i = x_i'$	$\underline{x}$	x_i'	$\bar{x}$	x_i'	$\bar{x}$	$\bar{x}$
	$x_i' \sim x_i^*$	$x_i^* < x_i'$	$x_i^* > x_i'$	$x_i^* > x_i'$	$x_i^* > x_i'$	$x_i^* \sim x_i'$

This is for median. for mean we better lie. (HW)

Iterated elimination

Dominance Solvable Game: (Gibbens)

For A if a_i dominates a_j (strictly) remove a_j completely

Maxi-Min (Dixit and Skeath)

Payoff of 1 is a_i ; then payoff of 2 is a_j
(zero-sum game)

	L	M	R	Min
T	2	5	13	2
M	6	5.6	10.5	5.6
L	6	4.5	1	1
B	10	3	-2	-2

Maxi-min = 5.6

Solution: (H, M)

Max	10	5.6	13
-----	----	-----	----

Mini-max = 5.6

Zero-sum Game $\Rightarrow$ Strictly Competitive Game.

Now,

	L	C	R	Min
T	3	-2	2	-2
M	-1	0	4	-1
B	-4	-3	1	-4
Max	3	0	4	

No solution.

Rationalisability:

Best response (equality allowed)

(Max-Cell)

	C	NC
C	1, 1	3, 0
NC	0, 3	2, 2

For 1 is C rationalisable

C is BR to C of 2

C " " " NC " 2

NC of 2 is rationalisable? No

C of 2 is rationalisable

	a	b
a	1, 0	0, 1
b	0, 1	1, 0

a is rationalisable w/ d

d " " " b

b " " " c

c " " " a

Symmetric Games and Symmetric Equilibrium:

$u_1(a_1, a_2) = u_2(a_2, a_1)$ $\rightarrow$ NE is steady state must be along the diagonal

Ex.

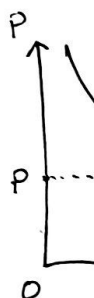
	x	y
x	c, c	a, b
y	b, a	d, d

So,

0, 0	2, 2
2, 2	0, 0

not symmetric

NE is not steady state



NE's (A,A) , (A,C) , (C,A)
Symmetric NE (A,A)

Oligopoly Markets:

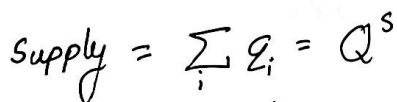
↓
Bertrand Model

Current Oligopoly: (model used when producer controls amount of supply)

- Now, $\pi_i(\cdot) = \text{Total Revenue} - \text{Total Cost}$

$P = \text{price}$
 $C_i(z_i) = \text{Cost for producing } z_i$
 $C_i'(z_i) > 0$

or, Based on the ~~sub~~ equilibrium of Supply and demand.



Now, Inverse demand f^n : $P = f^{-1}(Q^d)$... Now, $f(f^{-1}(Q)) = Q^d$

$$S_0, f^{-1}(a)^{-1} = \frac{1}{f'(f^{-1}(a))} < 0$$

$$s_0, \pi_i(\cdot) = f^{-1}(\sum_i z_i) \quad z_i = C_i(z_i)$$

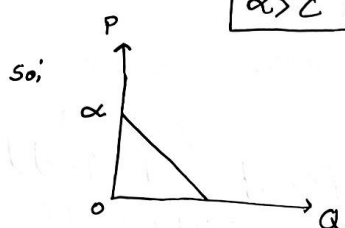
$$\text{So, } \pi_i(z_1, \dots, z_n) = f^{-1}(\sum z_i) * z_i - c_i(z_i)$$

Assumptions:

i) $n=2$; $C_i(Q_i) = cQ_i$; $c > 0$; c unit cost of production.

$$Q^d = \alpha - P \quad \alpha > 0 ; \text{ inverse dd } f^n: \quad P = \alpha - Q \quad \text{if } \alpha > Q$$

$$= 0 \quad \text{if } \alpha < Q$$



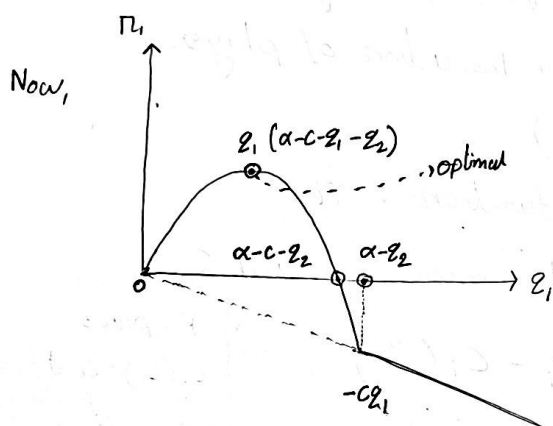
Now, $\pi_1(q_1, q_2) = P_{q_1} - c q_1$

$$= \frac{q_1 (P - c)}{2} = \frac{q_1 (\alpha - q_1 - q_2 - c)}{2}$$

$$= \frac{q_1 (\alpha - c - q_1 - q_2)}{2}$$

So, $\pi_1(q_1, q_2) = \frac{q_1 (\alpha - c - q_1 - q_2)}{2}$ if $\alpha > q_1 + q_2$

$$= -c q_1 \quad \text{if } \alpha < q_1 + q_2$$

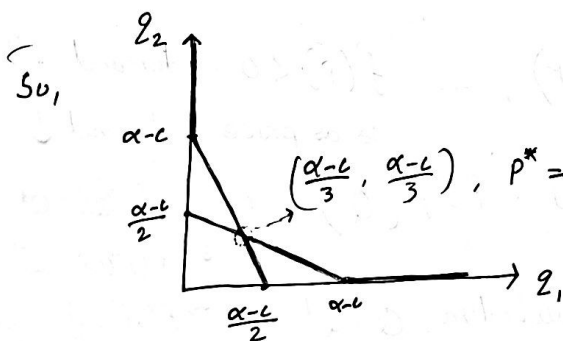


$$\left\{ \begin{array}{l} \text{So } \frac{\partial}{\partial q_1} \pi_1(q_1, q_2) = (\alpha - c - q_2) - 2q_1 \\ \text{So, } q_1 = \frac{\alpha - c - q_2}{2} \end{array} \right.$$

only for $q_2 \leq \alpha - c$

for $q_2 > \alpha - c$

So, $B_1(q_2) = \begin{cases} \frac{\alpha - c - q_2}{2} & \text{if } q_2 \leq \alpha - c \\ 0 & \text{if } q_2 > \alpha - c \end{cases}$



$$p^* = \frac{\alpha}{3} + \frac{2c}{3}, \quad \pi_1^* = \left(\frac{\alpha - c}{3} \right)^2$$

(demand) $\uparrow \Rightarrow q_1^* \uparrow, Q^* \uparrow, P^* \uparrow, \pi_1^* \uparrow$

Now $c \downarrow \Rightarrow q_1^* \uparrow, Q^* \uparrow, P^* \downarrow, \pi_1^* \uparrow$

(Technology) $\uparrow$

Now alternative 1) $C_1 \neq C_2$, 2) $C_i(C_i) = C_i^2$

Now, 2 firms. 1st firm maximize market share but profit ≥ 0
2nd " " Profit

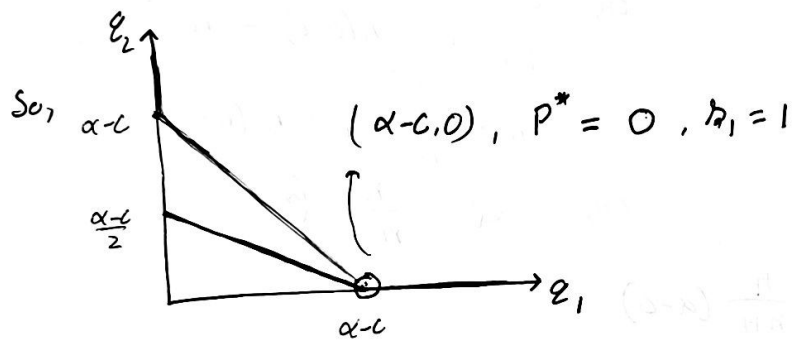
Now, $h_1(z_1, z_2) = \frac{z_1}{z_1 + z_2}$ and $\pi_1(z) \geq 0$

So, $h(z_1, z_2) \uparrow$ when $z_1 \uparrow$

so, untill $\pi_1(\cdot) = 0$ we will $\uparrow q_1$

$$\begin{aligned} \text{So, } B_1(c_2) &= z_1 \text{ at } [\pi_1(c) = 0] \\ &= \begin{cases} \alpha - c - z_2 & \text{if } \alpha - c \geq z_2 \\ 0 & \text{if } \alpha - c < z_2 \end{cases} \end{aligned}$$

$$B_2(z_1) = \begin{cases} \frac{\alpha - C - z_1}{2} & \alpha - C \geq z_1 \\ 0 & \alpha - C < z_1 \end{cases}$$

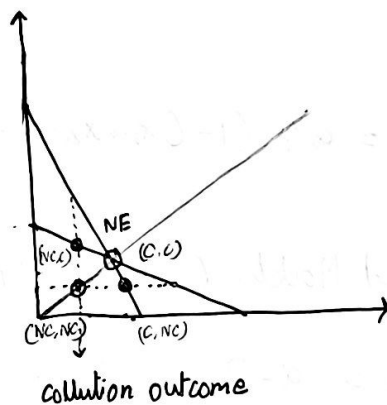


Collusion/ Carteling Anti Trust Laws bans carteling in some areas

Joint profit maximize ie more than one firm collaborate.

Eg. OPEC

$$\begin{aligned}\text{Now, } \pi &= \text{Joint profit} = \pi_1 + \pi_2 \\ &= Pq_1 - cq_1 + Pq_2 - cq_2 \\ &= (P-c)(q_1 + q_2) \\ &= (a-bQ-c)Q\end{aligned}$$

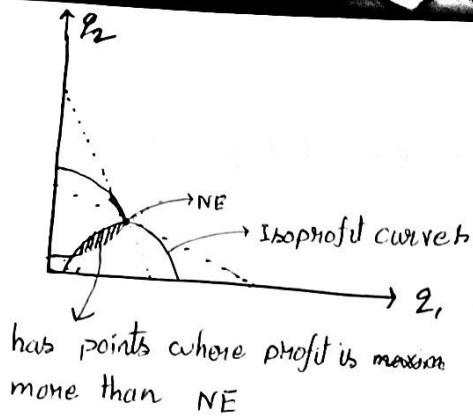


Now, $\frac{\partial \pi}{\partial Q} = 0 \Rightarrow (\alpha - c) - 2Q = 0$ or, $Q = \frac{\alpha - c}{2}$

$$\text{So, } q_1^* > \bar{q}_1 = \bar{q}_2 = \frac{\alpha - c}{4} \quad ; \quad \bar{p} = \frac{\alpha + c}{2} > \frac{\alpha + 2c}{3} = p^*$$

Now, $\bar{\pi}_i = \frac{\alpha-c}{9} \cdot \frac{\alpha-c}{2} = \frac{1}{18} (\alpha-c)^2 > \pi_i^* = \frac{1}{9} (\alpha-c)^2$

Not NE
what is the
profitable
deviation?



Say, $(z_1^*, \dots, z_n^*) = NE$

$$\begin{aligned} \text{So, } z_i &= (\alpha - c) - \frac{n}{n+1} (\alpha - c) \\ &= \frac{1}{n+1} (\alpha - c) \quad \forall i \end{aligned}$$

Now, $z^* = \frac{1}{n+1} (\alpha - c)$, $Q^* = \frac{n}{n+1} (\alpha - c)$

$$p^* = \frac{\alpha + nc}{n+1}$$

$$\pi^* = \left(\frac{\alpha - c}{n+1} \right)^2$$

Now, as $n \rightarrow \infty$ $p^* \rightarrow c$ [Perfect competition]

Q) $U_i = x_i (1 - (x_1 + x_2 + \dots + x_n))$ if $\sum x_j \leq 1$ x_i = resource used by i

Bertrand Model: (firm control price)

$$\left. \begin{aligned} D(P) &= \alpha - P & P \leq \alpha \\ &= 0 & \text{ow} \end{aligned} \right\} \text{Demand } f^n$$

$$C_i(z_i) = cz_i \quad \text{production}$$

$$\boxed{\alpha > c}$$

$$\pi_i(P_1, P_2, \dots, P_n) = TR - TC$$

Case I: I suggest a minimum price P_i with other n firms. But the other $n-m$ firms have quoted price $> P_i$:

$$\text{so, } \pi_i(P_1, P_2, \dots, P_n) = P_i \frac{D(P_i)}{m} - c_i \left(\frac{D(P_i)}{m} \right)$$

Now, for $n \geq 2$

$$\text{so, } P = \alpha - Q, \quad C_i(z_i) = cz_i$$

$$\alpha > c > 0$$

$$\begin{aligned} \pi_i(z_1, z_2, \dots, z_n) &= Pz_i - cz_i \\ &= z_i \left(\alpha - \left(\sum_{j=1}^n z_j \right) - c \right) \end{aligned}$$

$$\frac{\partial \pi_i}{\partial z_i} = \alpha - \sum_{j=1}^n z_j - c - z_i = 0$$

$$\Rightarrow B_i(z_1, \dots, z_n) = \alpha - c - \sum_{j=1}^n z_j - z_i$$

$$\Rightarrow z_i = \alpha - c - \sum_{j=1}^n z_j$$

$$\text{oh, } \sum_{i=1}^n z_i = n(\alpha - c) - n \sum_{j=1}^n z_j$$

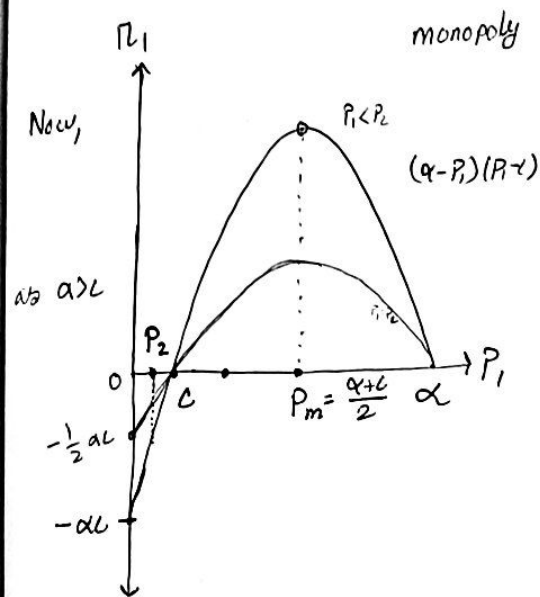
$$\text{oh, } (n+1) \sum_{i=1}^n z_i = n(\alpha - c)$$

$$\text{oh, } Q = \frac{n}{n+1} (\alpha - c)$$

Case I: I am counting price that is not min
 $\pi_i(p_1, \dots, p_n) = 0$

Now, First Duopoly.

① $n=2$, ② $p_i \geq 0$ [action] ③ $\pi_1(p_1, p_2) = \begin{cases} p_1(\alpha - p_1) - c(\alpha - p_1) & \text{if } p_1 < p_2 \\ = (\alpha - p_1)(p_1 - c) \\ = \frac{1}{2}(\alpha - p_1)(p_1 - c) & \text{if } p_1 = p_2 \\ 0 & \text{if } p_1 > p_2 \end{cases}$



Now, $B_1(p_2) = ?$

- Case I: $p_2 < c \dots B_1(p_2) = \{p_1 : p_1 > p_2\}$
Case II: $p_2 = c \dots B_1(p_2) = \{p_1 : p_1 \geq p_2\}$
Case III: $c < p_2 \leq P_m \dots B_1(p_2) = \emptyset$ [no optimal] as continuous
Case IV: $p_2 > P_m \dots B_1(p_2) = P_m$ for Discrete there is